# Statistical Learning and Data Analysis
## Prolusion

**Roberta Siciliano**

*Statistics, tèchnè-loghìa, analysis of data*

**DIETI, University of Napoli Federico II**

# Statistical Learning and Data Analysis: two modules and one final exam (cfu 12)

Module A (cfu 6):
**Statistics: the foundations**

Prof. Emiliano del Gobbo

Module B (cfu 6):
**Statistical Learning**

Prof. Roberta Siciliano

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# Statistical Learning and Data Analysis – cfu 12

## Module A – Foundations of Statistics for Data Scientists

- ✓ Exploratory Data Analysis, Probability and Inference, Linear Models

## Module B – Statistical Learning

- ✓ Dimensionality Data Reduction, Modelling and Prediction

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# Aim of the Course

- *"Education is not the piling on of learning, information, data, facts, skills, or abilities – that's training or instruction – but is rather making visible what is hidden as a seed."*— Thomas More, Irish writer, poet and lyricist.
- Aim of the Course of **Statistical Learning and Data Analysis** is to transfer a **statistical education** and a **rigorous methodological approach** such to be effective and efficient in transforming data into useful information which provides an added value or outcome for knowledge and decision-making.
- Acquire the ***operational steps*** from data preparation to the evaluation of results, and to develop practical skills in the generation, analysis and interpretation of results through practical exercises carried out with open-source tools.

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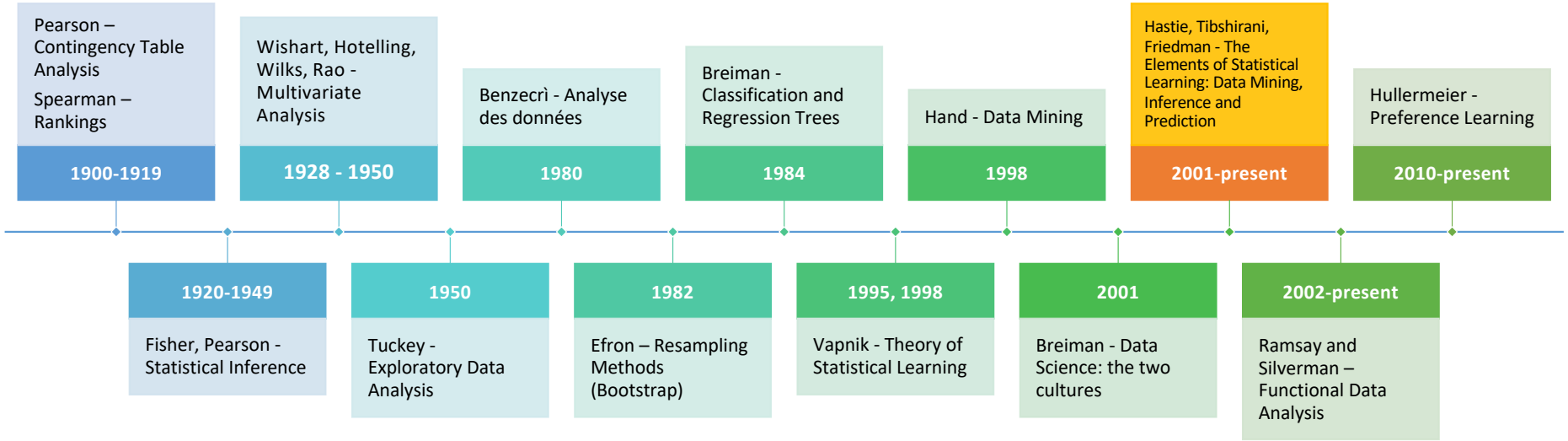
# Historical statistical flashback toward **Statistical Learning, Functional Data Analysis and Preference Learning**

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# Textbooks

- **Foundations of Statistics for Data Scientists,** with R and Python, Alan Agresti and Maria Kateri, CRC Press (2022)
- **Introduction to Statistical Learning, with applications in R**. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani (2009) (Available for free in pdf)
- **Introduction to Statistical Learning, with applications in Python**. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Jonathan Taylor (2023) (Available for free in pdf)
- **The elements of Statistical Learning: Data Mining, Inference, and Prediction**. Hastie, T., Tibshirani, R., Friedman, J., Springer (2009) (https://hastie.su.domains/ElemStatLearn/)

- https://www.statlearning.com to download ISL with R as well as with Python, resources (code, slides, data sets, figures, et.)

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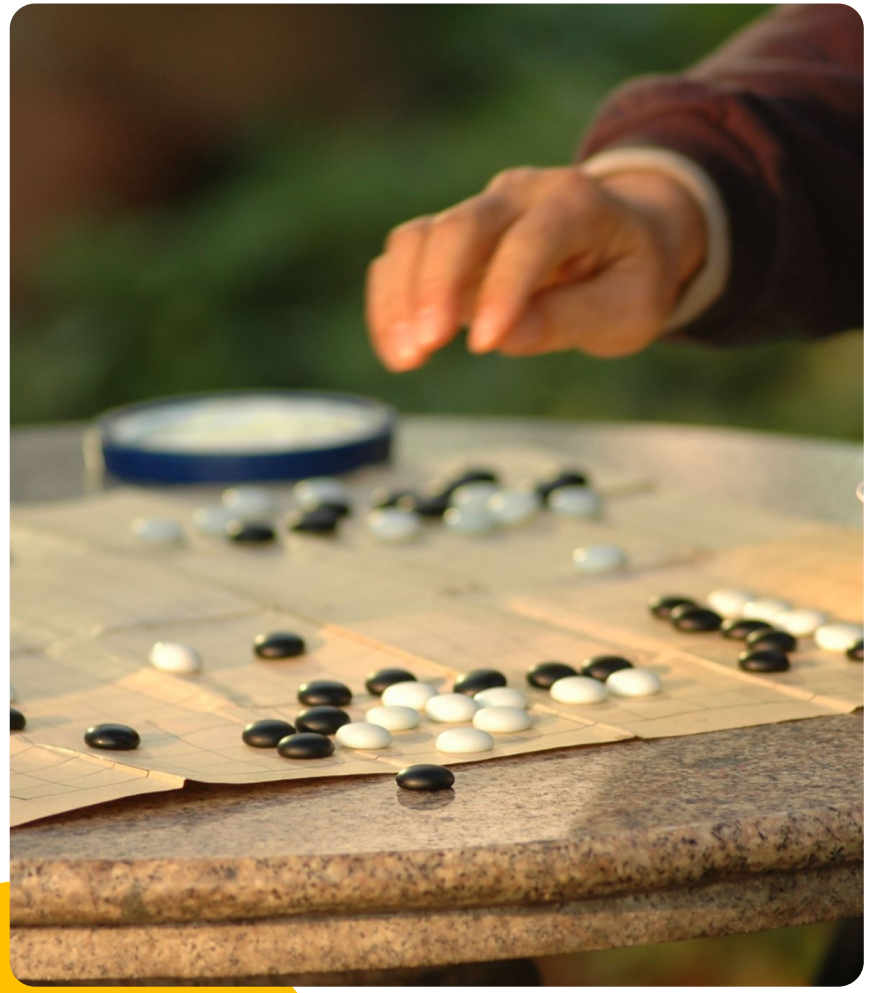

# A classical thought

*«Statistics is a two-person game in which the first is the mathematician and the second is nature»*

*Abraham Wald* (1939)

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# A modern thought

*“Vast amounts of data are being generated in many fields, and the statisticians job is to make sense of it all: to extract important patterns and trends, and understand what the data says. We call this learning from data"* *(Hastie, Tibshirani, Friedman, 2001, 2009)*

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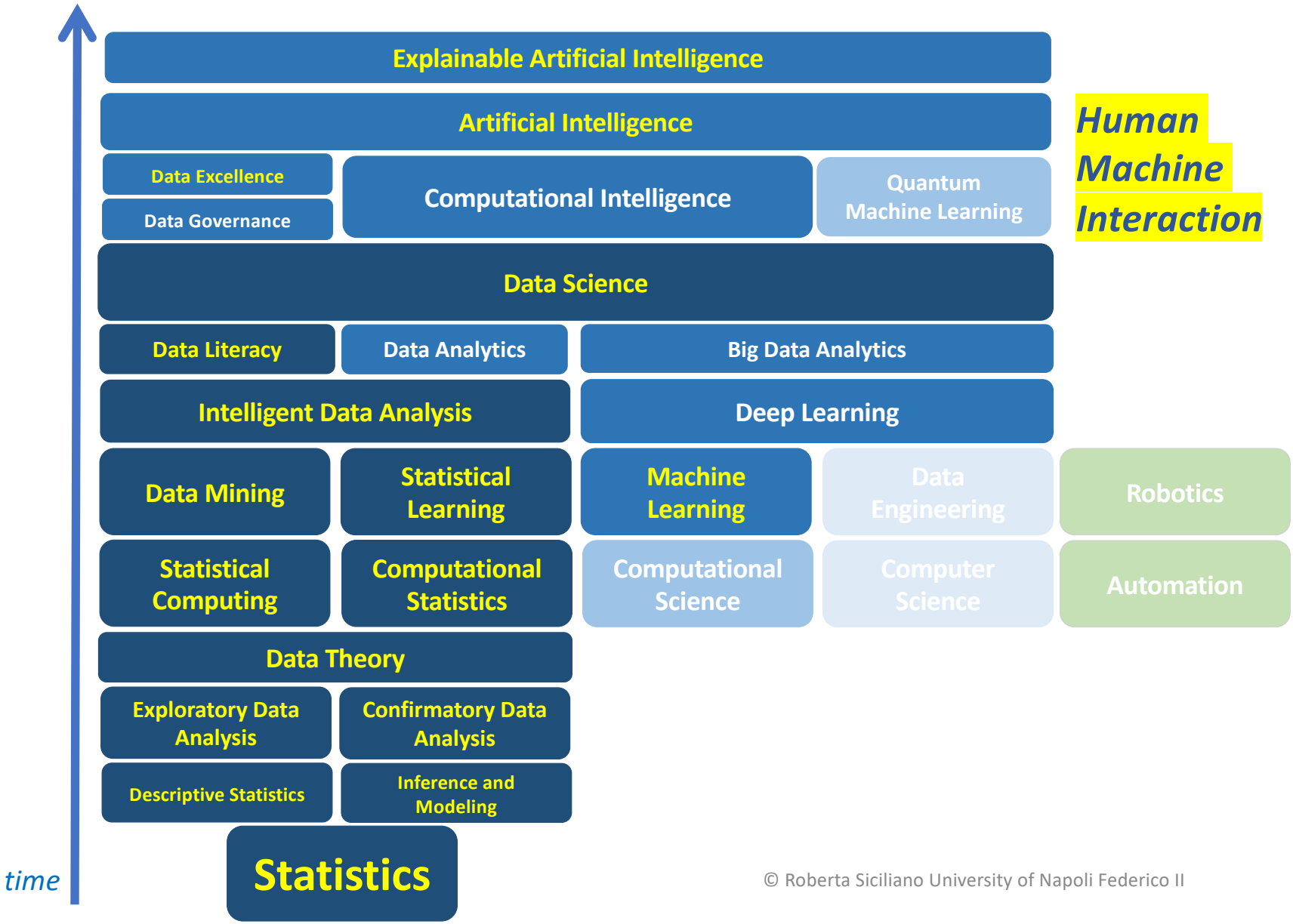


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# The Human Side of Science

Data are everywhere as also Artificial Intelligence

Intelligence is not everywhere

***"The most valuable intelligence is not artificial."***

There are always humans in science and applications.

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# The beginning of my scientific story

- Centro di Specializzazione in Ricerche Economico-Agrarie per il Mezzogiorno di Portici, **Advanced Statistics Master**, coordinator **Domenico Piccolo** (Textbook in Italian: Statistica, Il Mulino).
- **PhD in Computational Statistics and Applications** of Department of Mathematics and Statistics, University of Naples Federico II, coordinator **Carlo N. Lauro**, mentore of the **LAMDA (Laboratorio per l'Analisi Multidimensionale dei Dati)** as followers of **Jean Paul Benzecrì** with ***Analyse des Donnèes*** in France and **Chikio Hayashi** in Japan with the **Quantifications of qualitative data**, one top scientist and **founder of Data Science**

  From Wikipedia: Carlo Lauro (past president of the International Association for Statistical Computing) defined
  - **'statistical computing'** as *"the application of computer science to statistics"*,
  - **'computational statistics'** as *"aiming at the design of algorithm for implementing statistical methods on computers, including the ones unthinkable before the computer age (e.g. bootstrap, simulation), as well as to cope with analytically intractable problems"*
- Leiden University (Technieken en Methoden Institute) in the Netherlands, **visiting research periods**, supervisors **Peter GM van der Heijden**, **Ab Mooijaart**; GIFI School of **Jan de Leeuw** (Optimal Scaling), **Data Theory Group** coordinated by Willem Heiser and **Jaqueline Meulman**
  - **Mark de Rooij**, student of Willem Heiser, is now Professor of Artificial Intelligence and Data Theory in the Ms Statistics and Data Science, Leiden University **[ERASMUS Agreement with Leiden University]**
- Two key issues were soon clear to me:
  - there is no need of Statistics without an application domain with a real-world question requiring data analysis;
  - Computational Statistics is Statistics in the Computer Era (definition of **Jerome Friedman**).

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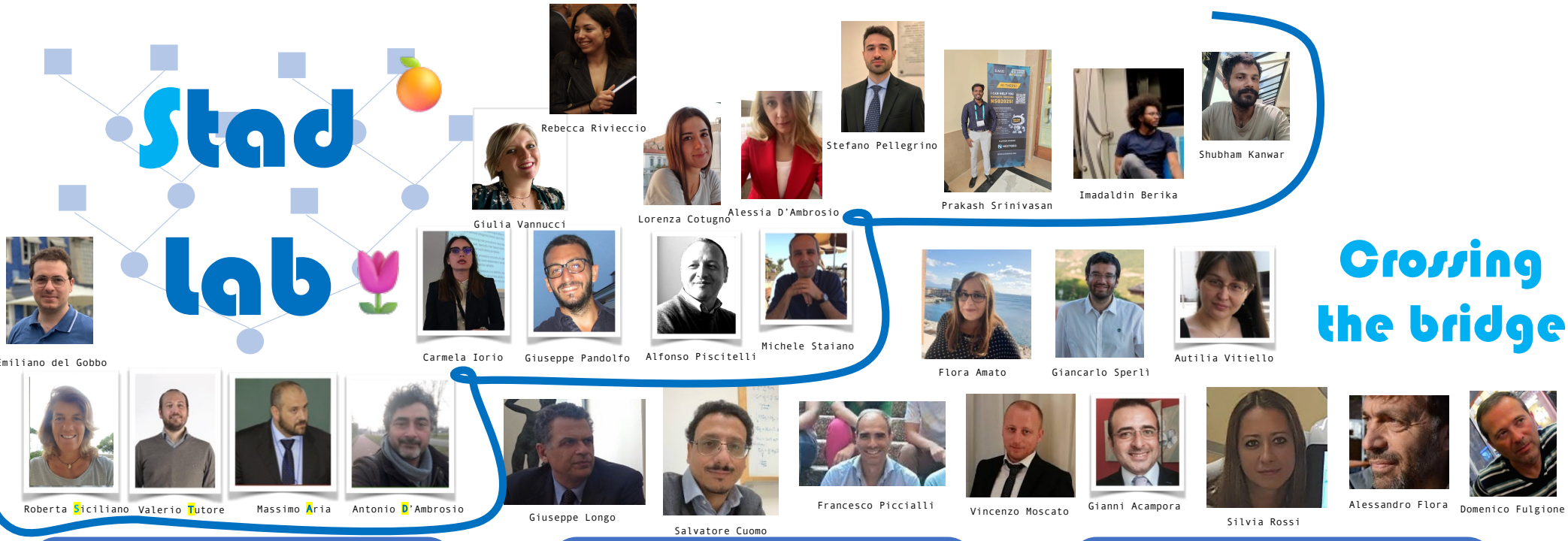
# Stad lab




## Crossing the bridge

STAD is "city" in Dutch - tribute to the founders' background acquainted at the Leiden University in the Netherlands which countryside was flat long before the globalization.

[Statistics, technology, analysis of data] is a "research city" of University of Naples Federico II in the flat world where a new "citizen" can aggregate from almost anywhere.

This city contains a lot of "buildings" to create and work out ideas, provide statistical science, transfer technology, analyze data with a meta-disciplinary approach, "crossing the bridge" towards other scientific communities.

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# About Stad lab

**Stad = city in Dutch**

Orange

Country of Tulips

- **Statistics, *tèchnè-loghìa*, analysis of data** are key words for reading the reality as it is manifested in **qualitative** expressions (categories, attributes, classes, groups, labels, etc.) and **quantitative** ones (numbers, measurements, values, etc.) in various domain contexts or application fields.
- All starts with a real-world case study and specific questions that require:

✓**Statistical Learning & Data Analysis** for standard datasets

✓**Data Mining & Machine Learning** for complex data structures

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# **Stad** – The etymological derivation

- **Statistics** from the Latin **Status** = what it is, **how to perceive the reality**, from the German "Stadt" = city (*"science dealing with data about the condition of a state or community",* by the German political scientist Gottfried Aschenwall, in "Vorbereitung zur Staatswissenschaft", 1748).
- ***tèchnè-loghìa*** combines two Greek words:
  - ***tèchnè*** *= art, concrete ability of doing, the human brain to 'remember', 'think', 'come to mind', 'cross the mind', 'invent something new',*
  - ***loghìa*** *= discourse, reasoning on art, explanation, knowing how to do*
  - ***tèchnè-loghìa*** = **the concrete ability and reasoning on how to do**
- **analysis** = **solution of a problem** (Latin), *"the process of breaking a complex topic into smaller parts in order to gain a better understanding of it" (Aristotele, 384-322 B.C.),*
- **data** = **known quantities** (Latin).

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# Statistics, *tèchnè-loghìa*, analysis of data

- **Statistics** has two fundamental missions:
  - ✓ **Exploratory Data Analysis** *(from data to information)* uses a *"deductive approach"* to discover significant facts, patterns, groups, anomalies, associations, correlation, similarities, typologies, clusters, and so on;
  - ✓ **Confirmatory Data Analysis** *(from information to knowledge)* uses an *"inductive approach"* to justify theories and hypotheses, to build up models for decision-making and prediction.
- ***tèchnè-loghìa*** is the rationalization process that starting from the real-world questions build up a "project" to move from theory to practice. The concrete ability pass through two attitudes:
  - ✓ ***Heuristic Experience*** through trial and error to seek solutions for a new problem,
  - ✓ ***Algorithmic Experience*** by applying known solutions for a problem that is very similar to others previously addressed.
- **analysis of data** is to find solutions from known quantities, the data.

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# Raw data

- **Data is the cornerstone of Statistics.**
- **Data** results from the measurement of a *variable* (or feature) on a *unit* (or object, individual) under observation or experimentation at a given time and space.
- The **size** of a data set refers to the number of units whereas the **dimensionality** to the number of variables. Typically, **multidimensional data** are analyzed.
- On the basis of the scale of measurement, we distinguish between ***quantitative (or numerical) variables*** and ***qualitative (or categorical) variables***. A further distinction is between ***ordinal*** and ***nominal*** variables if the categories of the qualitative variable can be ordered or not.
- ***Coding*** is the recording with a *"value"* the result of the measurement so that the "value" can have *an intrinsic value number* that can be processed *for quantitative variables* or is just *pointing out a specific category for a qualitative variable and then suited for classification of data points.*
- **Raw data** can be understood as a row vector or **record of values** *corresponding to the measurements of all variables on the unit.*

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# What do we mean for data?

- **We live in a world of data of many different types!** All starts with data and a **real-world case study in a domain**.
- Data can be extracted by various **data sources** and are of ***different types:***
  - **Surveys:** standard data can be collected directly and organized in a data matrix units x variables.
  - **Data Warehouse:** the store of ***operational data*** collected and stored by firms, private and public institutions for the proper functioning of their management activities.
  - **Sensors, Internet of Things devices, Tracking devices and other smart devices** collect ***data stream*** (a sequence of digitally encoded coherent signals, often organized into a series of "packets").
  - **Satellites** collect images and data using cameras.
  - Web portals collects ***web log data*** on the web navigation on internet.
  - **Audio-visive technological devices** collects ***audio data***.
  - **Social Networks** provides ***text data*** and ***image data*** as well as ***graph data***.
  - **Electrocardiography Sensor (ECG)** extract electrical signals in the heart that can summarized as ***time series*** as well as some text data and standard data types associated to the patient information.
  - **Geographic maps** provides ***spatial data.***
- **Non-standard data types can be transformed into standard data type in data pre-processing step.**

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# Key concepts

Raw data and their *mutability* is the *"heart"* of Statistics.

The output of data analysis can be any form of summarization and visualization that need *to be interpreted correctly*.

Statistical Data Analysis is *effective* if it provides an outcome in terms of innovation, satisfaction, utility, revenue, consequential actions etc. The *timeliness* plays a key role for useful decision-making and prediction.

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# Statistical Learning and Data Analysis: the mission

**The statistician or data scientist**, to understand the facts (exploration) and to confirm the theories (confirmation), makes use of *reasoning and concrete skills* (statistical thinking and methodology) to make the most of the available data with the related domain context information as a result of observing reality.

The final goal of **Statistical Learning and Data Analysis** is *to learn from data* and *find solutions* to a real-world problem.

All starts with real-world case study in a domain context.

The statistician or data scientist needs to transform ***raw data (input)*** into ***useful information and knowledge (output)*** (tables, plots and graphs, info-graphics, statistical indexes, models, prediction) to be correctly interpreted and analyzed such to provide answers to the real-world questions as well as some ***actions and consequences (outcome) to add value*** (utility, revenue, satisfaction, innovation plan, etc.) for some stakeholders.

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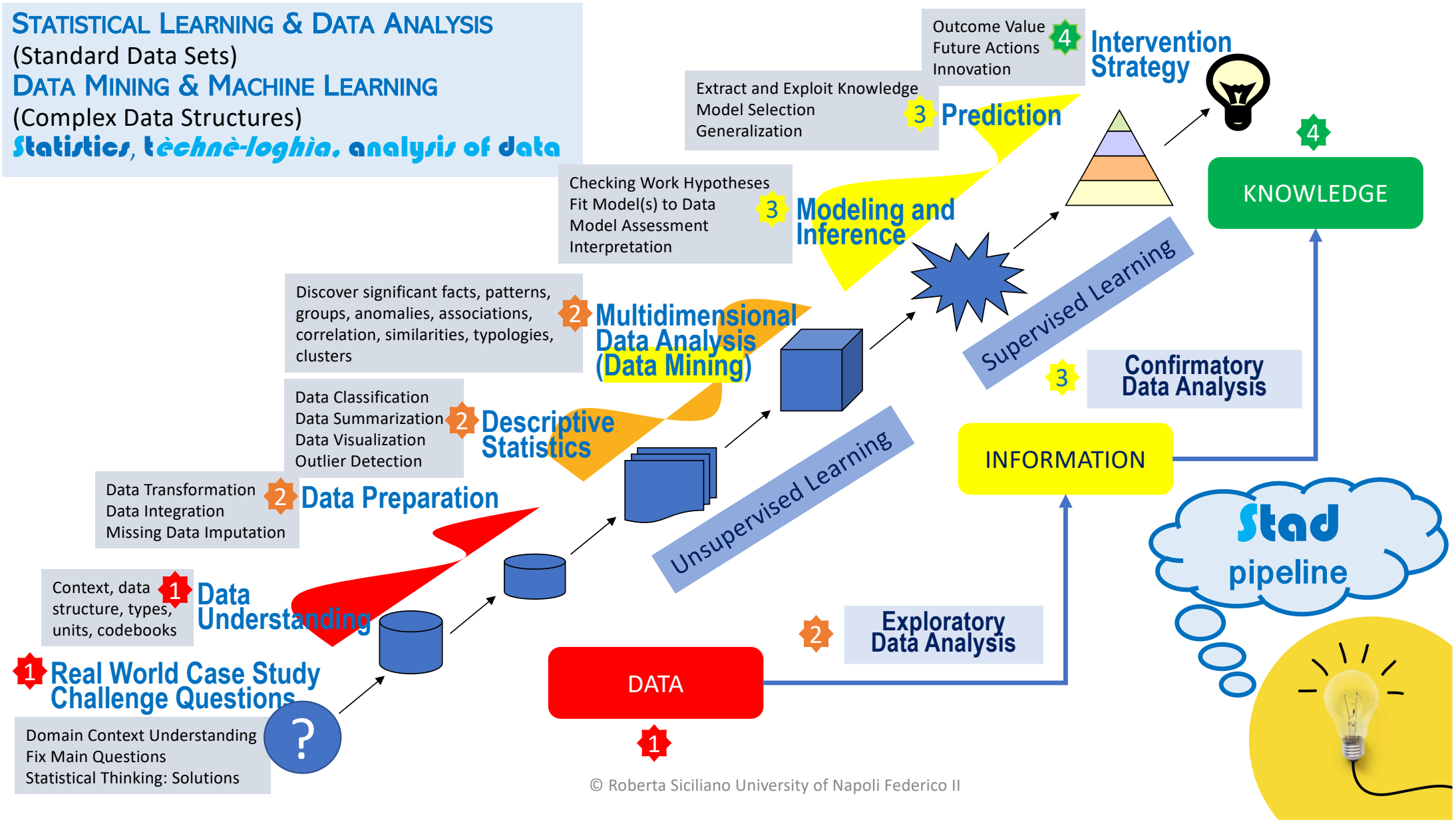


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# Technical Report: *sessions and contents structure*

## 1 Case Study Challenge

Introduce the Domain Context and the Real-World Case Study: **domain understanding** and which **challenging questions to satisfy**.

Refer to **the link of the databases**.

**Data Understanding**: describe the context, the data structure, the typology of variables, the units, time/space, etc.

**Statistical thinking**: convert questions into statistical problems, summarize and motivate which statistical methods to apply.

## 2 Exploratory Data Analysis

**Data Preparation:** Data pre-processing and cleaning, data transformation, data integration, missing data imputation.

**Descriptive Statistics:** data summarization, data visualization outlier detection.

**Multidimensional Data Analysis (basic Data Mining strategy):** dimensionality data reduction and clustering.

**Statistical thinking:** Summarize the relevant information derived from the Exploratory Data Analysis.

## 3 Confirmatory Data Analysis

**Modeling and Inference:** Checking Work Hypotheses, Fit Model(s) to Data, Model Assessment, Interpretation.

**Prediction:** Extract and Exploit Knowledge, Model Selection and Generalization.

**Statistical thinking:** Summarize the knowledge extraction due to the Confirmatory Data Analysis.

## 4 Outcome and concluding remarks

**Summarize the key points of the outcome of the Statistical Data Analysis.**

**Intervention Strategy:** Outcome Value, Future Actions, Innovation, how the beneficiary/stakeholders utilize the outcome of statistical data analysis.

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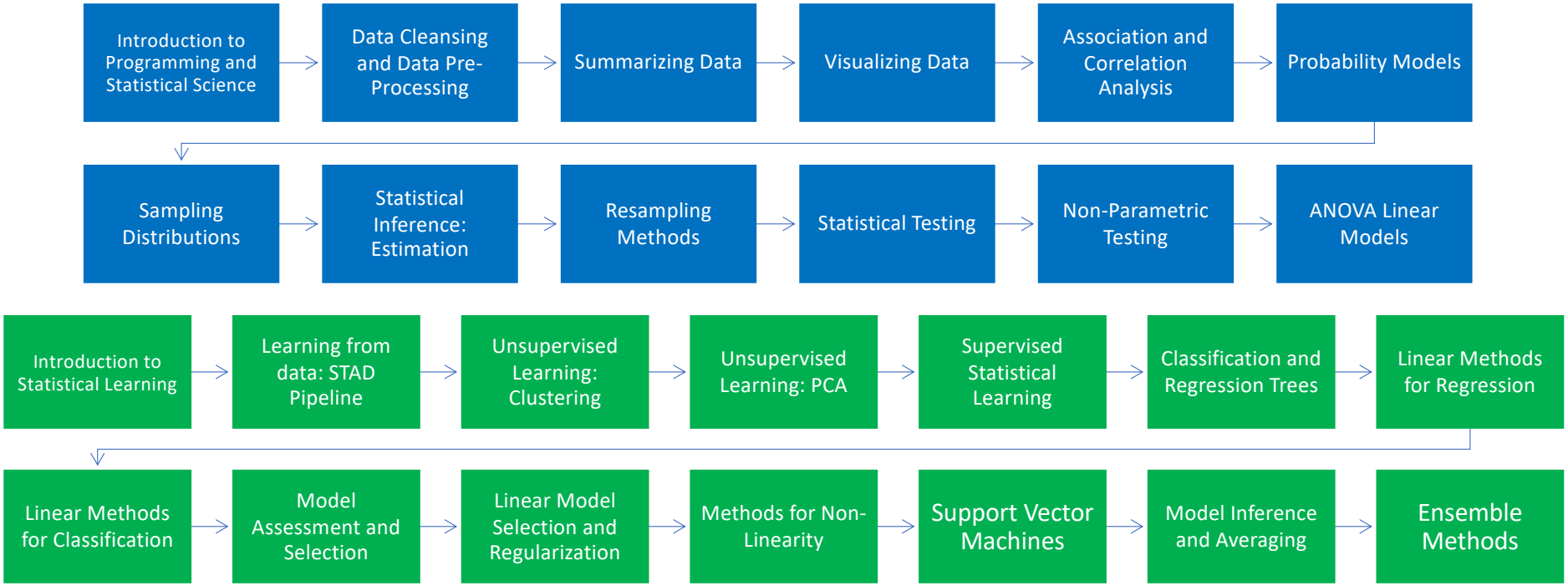
# Lectures Road Map Modules A and B

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# Laboratory in R

✓https://cran.r-project.org

✓https://rstudio.com

✓https://posit.cloud/

→a cloud-based version of Rstudio

✓https://swirlstats.com

→ a tutorial

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# Laboratory in Python

- https://www.python.org
- Access to Jupyter, a very popular Python interface that runs code through a file called a *notebook*.
- Download and install Python3 by following the instructions available at anaconda.com.
- There are a number of ways to get access to Jupyter.
- Here are just a few:
  - Using Google's Colaboratory service: colab.research.google.com/.
  - Using JupyterHub, available at jupyter.org/hub.
  - Using your own jupyter installation. Installation instructions are available at jupyter.org/install.

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# Data sets

- ISLP Package and Data sets
- Kaggle Repository https://www.kaggle.com/datasets
- UCI Machine Learning Repository https://archive.ics.uci.edu/ml/index.php
- Others (i.e., Football, …)

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# Exam

We ask you to build your **Portfolio of Statistics Projects** by responding to assignments to do at home or in class during the Course.

At the Exam, we evaluate your **Portfolio of Statistics Projects**.

You will have to answer methodological, computational and application questions.

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# The Exam Grade: the evaluation grid

- The Exam Grade considers the following evaluation grid:

| Criterion<br>Exam Grade | Knowledge of the contents | Rigor in applying concepts and methods | Quality of the Reporting and the Script | Clarity of the Exposure |
|---|---|---|---|---|
| **< 18 Not sufficient** | *fragmented and superficial* | *fundamental errors* | *insufficient* | *insufficient* |
| **18 - 20** | *sufficient but general concepts* | *uncertainties in the application of concepts and methods* | *simplified reporting and sufficient script* | *elementary exposure* |
| **21, 22, 23, 24** | *appropriate but not in-depth knowledge of contents* | *ability to apply fundamental concepts* | *ability of reporting in a simple way and good script* | *clear and correct presentation* |
| **25, 26, 27** | *precise and complete knowledge of contents* | *ability to apply broad and complete concepts* | *ability of reporting in an articulated way and very good script* | *safe and correct exposure* |
| **28, 29, 30, 30 et lauda** | *very broad, complete, and in-depth knowledge of contents* | *well-established ability to apply contents and excellent ability to analyze, synthesize and interdisciplinary connections* | *ability to present broad, complete and in-depth contents in an articulated way and excellent script* | *mastery of exposure* |

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